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N-body Simulations of Spiral Bending Waves in Planetary Rings

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ABSTRACT

Draft. This mock paper currently holds the derivation of the vertical epicyclic motion used by the `epi_int_3d` drift step (Section 2), the vertical gravity that one streamline exerts on another (Section 3), its relation to `epi_int_lite`'s in-plane streamline gravity (Appendix A), and a study of the streamline method's resolution limit (Appendix B), and the vertical restoring force of a bending wave (Appendix C), and the theory of spiral bending waves at a vertical resonance with its tests against spiral wave theory (Section 4): the forced response of massless particles, and the bending wave that the satellite launches in a self-gravitating ring, which tends to the massless particles' response as the ring's surface density goes to zero (Appendix D).

1. INTRODUCTION

The `epi_int` and `epi_int_lite` N-body integrators (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025) model a planetary ring as a set of gravitating streamlines whose particles are confined to the ring plane. Spiral bending waves, which are launched at a satellite's vertical resonances (F. H. Shu et al. 1983; F. H. Shu 1984), require the particles' vertical motions too. This paper describes `epi_int_3d`, a modest extension of `epi_int_lite` into the vertical (out-of-plane) direction, and uses it to simulate spiral bending waves.

2. VERTICAL EPICYCLIC MOTION

The `epi_int_lite` drift step advances each particle's unperturbed orbit about an oblate planet using first-order epicyclic equations that are exactly reversible: converting spatial coordinates into epicyclic orbit elements and back introduces no error, so the drift step is not a source of numerical drift (J. M. Hahn et al. 2025). Here those equations are extended to particles having vertical displacement z and vertical velocity v_z, with the same first-order accuracy and the same exact reversibility.

2.1. *Equations of motion*

A particle orbiting an oblate planet obeys Newton's second law,

$$\ddot{\mathbf{r}} = -\nabla\Phi_p, \tag{1}$$

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where Φ_p is the planet's gravitational potential. That potential is a series in the planet's zonal harmonics J_{2k} (Eqn. 5 of N. Borderies-Rappaport & P.-Y. Longaretti 1994), and as in J. M. Hahn & J. N. Spitale (2013) and J. M. Hahn et al. (2025) only the J_2 term is kept, so

$$\Phi_p(r,z) = -\frac{GM_p}{\rho}\left[1 - J_2\left(\frac{R_p}{\rho}\right)^2 P_2\left(\frac{z}{\rho}\right)\right]. \tag{2}$$

Here M_p is the planet's mass, R_p its radius, r the particle's cylindrical radius, $\rho = \sqrt{r^2+z^2}$ the particle's distance from the planet's center, and $P_2(z/\rho) = (3z^2/\rho^2 - 1)/2$ is the second Legendre polynomial. In cylindrical coordinates (r,θ,z), Eqn. (1) is

$$\ddot{r} - r\dot{\theta}^2 = -\frac{\partial\Phi_p}{\partial r} \tag{3a}$$

$$\frac{d}{dt}\left(r^2\dot{\theta}\right) = -\frac{\partial\Phi_p}{\partial\theta} = 0 \tag{3b}$$

$$\ddot{z} = -\frac{\partial\Phi_p}{\partial z}. \tag{3c}$$

The potential is independent of θ, so the z-component of the specific angular momentum, $h_z = r^2\dot{\theta} = rv_\theta$, is conserved exactly. Because the potential is even in z, near the ring plane

$$\Phi_p(r,z) \simeq \Phi_p(r,0) + \frac{1}{2}\nu^2(r)z^2, \tag{4}$$

where terms of order z^4 are neglected, $\Phi_p(r,0) = -GM_p/r - J_2GM_pR_p^2/2r^3$ is the midplane potential and

$$\nu^2(r) = \left.\frac{\partial^2\Phi_p}{\partial z^2}\right|_{z=0} = \frac{GM_p}{r^3}\left[1 + \frac{9}{2}J_2\left(\frac{R_p}{r}\right)^2\right]. \tag{5}$$

Equations (3) then become

$$\ddot{r} \simeq \frac{h_z^2}{r^3} - \left.\frac{\partial \Phi_p}{\partial r}\right|_{z=0} \tag{6a}$$

$$\ddot{z} \simeq -\nu^2(r) z, \tag{6b}$$

which neglect terms of order z^2 in the radial equation and z^3 in the vertical one.

So an unperturbed particle's horizontal and vertical motions are decoupled to first order in its small departures from a circular orbit in the ring plane. The ring's internal forces, its self-gravity, viscosity and pressure, can re-couple them. The vertical equation is that of a harmonic oscillator, so $\nu(r)$ is the frequency at which a particle at radius r oscillates vertically.

2.2. *Unperturbed motion*

J. M. Hahn et al. (2025) solved the planar equation of motion, Eqn. (6a), to first order in small displacements from circular motion, taking care to make the transformation from the coordinates r, θ, v_r, v_θ to the epicyclic elements $a, e, \tilde{\omega}, M$ exactly reversible, so that it can be used in a drift-kick N-body integrator like the one used here. That solution (Appendix A of J. M. Hahn et al. 2025) is summarized here and then extended to $|z| > 0$.

The particle's coordinates are

$$r = \frac{a}{\sqrt{1 + 2e\cos M}} \tag{7a}$$

$$\theta = \frac{\Omega_a}{\kappa_a}(M + 2e \sin M) + \tilde{\omega} \tag{7b}$$

$$v_r = ea\kappa_a \sin M \left(\frac{r}{a}\right)^3 \tag{7c}$$

$$v_\theta = \frac{a^2 \Omega_a}{r}, \tag{7d}$$

when written in terms of the epicyclic elements (Eqn. A9 of J. M. Hahn et al. 2025), where the mean angular velocity and epicyclic frequency at semimajor axis a are

$$\Omega_a = \sqrt{\frac{GM_p}{a^3}} \left[1 + \frac{3}{2} J_2 \left(\frac{R_p}{a}\right)^2\right]^{1/2} \tag{8a}$$

$$\kappa_a = \sqrt{\frac{GM_p}{a^3}} \left[1 - \frac{3}{2} J_2 \left(\frac{R_p}{a}\right)^2\right]^{1/2}. \tag{8b}$$

Equations (7) conserve the vertical component of the particle's specific angular momentum vector, $h_z = r v_\theta = a^2 \Omega_a$, and also solve the planar equation of motion with errors of order $\mathcal{O}(e^2)$. The particle's cylindrical radius, the length of its position vector projected onto the ring plane, is $r = a[1 + \mathcal{O}(e)]$. Its mean anomaly advances with time t at the epicyclic frequency, $M = \kappa_a t + M_0$, while a, e and $\tilde{\omega}$ are constant.

Equation (6b) is already first order in the small quantities, so $\nu(r) \simeq \nu(a)$ and the vertical oscillation frequency becomes

$$\nu_a \equiv \nu(a) = \sqrt{\frac{GM_p}{a^3}} \left[1 + \frac{9}{2} J_2 \left(\frac{R_p}{a}\right)^2\right]^{1/2}, \tag{9}$$

(Eqn. 17 of N. Borderies-Rappaport & P.-Y. Longaretti 1994), where the neglected terms are of order $\mathcal{O}(ei)$ and $\mathcal{O}(i^3)$. Outside the planet its potential obeys Laplace's equation, which for an axisymmetric potential is

$$\nabla^2 \Phi_p = \frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial \Phi_p}{\partial r}\right) + \frac{\partial^2 \Phi_p}{\partial z^2} = 0. \tag{10}$$

Note that $\Omega_a^2 = r^{-1} \partial \Phi_p / \partial r$, $\kappa_a^2 = 3\Omega_a^2 + \partial^2 \Phi_p / \partial r^2$ and $\nu_a^2 = \partial^2 \Phi_p / \partial z^2$, so Eqn. (10) relates the three frequencies via $\kappa_a^2 + \nu_a^2 = 2\Omega_a^2$.

The solution of Eqn. (6b) with $\nu = \nu_a$ is

$$z = a\, i \sin(\theta - \Omega) \tag{11a}$$

$$v_z = a \nu_a i \cos(\theta - \Omega), \tag{11b}$$

whose amplitude and phase define two new orbit elements, the inclination i and the longitude of ascending node Ω. The node regresses at the rate

$$\frac{d\Omega}{dt} = \Omega_a - \nu_a \simeq -\frac{3}{2} J_2 \left(\frac{R_p}{a}\right)^2 \Omega_a. \tag{12}$$

Also keep in mind that Ω is the longitude of ascending node, while Ω_a is the mean angular velocity. Equations (11) neglect terms of order $\mathcal{O}(ei)$, as Eqn. (9) does.

2.3. *Coordinates to elements*

Equations (7) and (11) can be inverted exactly via five steps, with the first four from Appendix A of J. M. Hahn et al. (2025):

1. Compute $h_z = r v_\theta$, then solve $h_z^2 = a^4 \Omega_a^2$ for the semimajor axis, $a = R_p \left(C + \sqrt{C^2 - 3J_2/2}\right)$ where $C = h_z^2 / 2GM_p R_p$.

2. Compute Ω_a and κ_a from Eqns. (8).

3. Compute $e \sin M = (v_r / a\kappa_a)(a/r)^3$ and $e \cos M = \frac{1}{2}[(a/r)^2 - 1]$, which give $e = \sqrt{(e \sin M)^2 + (e \cos M)^2}$ and $M = \mathrm{atan2}(e \sin M, e \cos M)$.

4. Compute $\tilde{\omega} = \theta - (\Omega_a / \kappa_a)(M + 2e \sin M)$.

5. The new fifth step: compute ν_a from Eqn. (9), then i and Ω from

$$i = \sqrt{\left(\frac{z}{a}\right)^2 + \left(\frac{v_z}{a\nu_a}\right)^2} \tag{13a}$$

$$\theta - \Omega = \text{atan2}(z, v_z/\nu_a). \tag{13b}$$

Once a is known, the mapping $(i, \theta-\Omega) \leftrightarrow (z, v_z/\nu_a)$ is a change between polar and Cartesian coordinates, so Eqns. (11) and (13) are exact inverses.

3. RING FORCES

A ring particle's epicyclic motion is also perturbed by the accelerations that the surrounding ring exerts on it: the ring's self-gravity, its pressure, and its viscosity, which accounts for the dissipative collisions that occur as particles in adjacent streamlines shear past one another. `Epi_int_3d` computes these accelerations as `epi_int_lite` does (Section 2 of J. M. Hahn et al. 2025). Its particles are assigned to streamlines, each a closed eccentric path around the planet that is populated by particles that share a common semimajor axis a and are distributed in longitude, so the particles trace out their streamlines. The particles do not interact with each other directly. Rather, each particle is perturbed by the accelerations exerted by the other streamlines, whose positions, slopes and linear densities near the particle are inferred from the particles that trace them.

3.1. *Gravity*

All `epi_int_3d` simulations span only a narrow annulus within a broader planetary ring, so every particle is radially close to the nearby portions of all perturbing streamlines. Which is why `epi_int_3d` and its predecessors (`epi_int` and `epi_int_lite`) assume that the gravitational acceleration of a particle by the particles inhabiting the perturbing streamline at semimajor axis a' is that due to a straight and infinitely long wire:

$$\mathbf{A}_g = \frac{2G\lambda'}{\Delta}\hat{\mathbf{\Delta}}, \tag{14}$$

where λ' is the perturbing streamline's linear density, Δ is the particle's distance from the streamline's nearest point, which is labeled the near point in Fig. 1, and $\hat{\mathbf{\Delta}} = \mathbf{\Delta}/\Delta$ is the unit vector that points from the particle toward the near point.

3.1.1. *The perturbing streamline's geometry*

The following 3D treatment is an extension of the 2D force calculations used in J. M. Hahn & J. N. Spitale (2013). In cylindrical coordinates the particle is at $\mathbf{r} = (r, \theta, z)$. The ring matter that is represented by the particles in perturbing streamline a' is at $\mathbf{r}' = (r', \theta', z')$ that traces a continuous wire, so these coordinates are to be regarded as functions of longitude θ', i.e., $\mathbf{r}' = (r'(\theta'), \theta', z'(\theta'))$. Lagrange polynomials will be fit to the particles in streamline a', which means that $r'(\theta')$, $z'(\theta')$ and the streamline's linear density $\lambda'(\theta')$ can be regarded as known functions in the following. And because the particle is close to the streamline, $|r'-r| \ll r$, $|z'-z| \ll r$ and $|\theta'-\theta| \ll 1$, so the particle's local frame $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\mathbf{z}})$ can be treated as Cartesian.

The vector from the particle to the streamline's near point (the black square in Fig. 1) can be written

$$\mathbf{\Delta} = \mathbf{r}'(\theta') - \mathbf{r} = \mathbf{d}_0 + \mathbf{d}_1, \tag{15}$$

where $\theta' = \theta + \delta\theta'$ is the near point's longitude as measured in the $z = 0$ plane, and $\delta\theta'$ is its longitude relative to the particle's with $|\delta\theta'| \ll 1$. This splits the trip to the near point across two legs. The first leg, $\mathbf{d}_0 = \mathbf{r}'(\theta) - \mathbf{r}$, travels from the particle to the crossing point $\mathbf{r}'(\theta) = (r'(\theta), \theta, z'(\theta))$, which is where the streamline crosses the particle's longitude (the cross in Fig. 1), so

$$\mathbf{d}_0 = d_r\hat{\mathbf{r}} + d_z\hat{\mathbf{z}}, \tag{16}$$

where $d_r = r'(\theta) - r$ and $d_z = z'(\theta) - z$. The second leg, $\mathbf{d}_1 = \mathbf{\Delta} - \mathbf{d}_0 = \mathbf{r}'(\theta') - \mathbf{r}'(\theta)$, then runs along the streamline from the crossing point to the near point. This is a right triangle (Fig. 1), so

$$|\mathbf{d}_0|^2 = \Delta^2 + |\mathbf{d}_1|^2. \tag{17}$$

That stretch of streamline is short and locally straight, so $\mathbf{d}_1 \simeq [r'(\theta') - r'(\theta)]\hat{\mathbf{r}} + r'(\theta)\,\delta\theta'\,\hat{\boldsymbol{\theta}} + [z'(\theta') - z'(\theta)]\hat{\mathbf{z}}$, which to first order in $\delta\theta'$ is

$$\mathbf{d}_1 \simeq (s_r\hat{\mathbf{r}} + \hat{\boldsymbol{\theta}} + s_z\hat{\mathbf{z}})\, r'\,\delta\theta' = Nr'\,\delta\theta'\,\hat{\mathbf{t}}, \tag{18}$$

where r' is short for $r'(\theta)$. This uses $r'(\theta') - r'(\theta) \simeq (dr'/d\theta')\delta\theta'$ and likewise for z', where the streamline's radial and vertical slopes are

$$s_r = \frac{1}{r'}\frac{dr'}{d\theta'}\bigg|_{\theta'=\theta} \quad \text{and} \quad s_z = \frac{1}{r'}\frac{dz'}{d\theta'}\bigg|_{\theta'=\theta}, \tag{19}$$

which are evaluated at the perturbed particle's longitude $\theta' = \theta$ and are obtained from the derivatives of the Lagrange polynomials. Together they give $s = (s_r^2 + s_z^2)^{1/2}$, which is the streamline's slope relative to circular motion in the ring plane, i.e., the $\hat{\boldsymbol{\theta}}$ direction, so the streamline is tilted from $\hat{\boldsymbol{\theta}}$ by the angle φ where $\tan\varphi = s$. Note that as $\delta\theta' \to 0$ the chord $\mathbf{d}_1$ points along the streamline, so $\hat{\mathbf{t}}$ is also the perturbing streamline's unit tangent:

$$\hat{\mathbf{t}} = \frac{s_r\hat{\mathbf{r}} + \hat{\boldsymbol{\theta}} + s_z\hat{\mathbf{z}}}{N}, \tag{20}$$

where $N^2 = 1 + s_r^2 + s_z^2$.

The near point is where $\mathbf{\Delta}$ is perpendicular to the streamline, $\mathbf{\Delta} \cdot \hat{\mathbf{t}} = (\mathbf{d}_0 + \mathbf{d}_1) \cdot \hat{\mathbf{t}} = 0$. And because $\mathbf{d}_1$ lies along $\hat{\mathbf{t}}$, this requires $\mathbf{d}_1 = -(\mathbf{d}_0 \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}$, so

$$\mathbf{\Delta} = \mathbf{d}_0 + \mathbf{d}_1 = \mathbf{d}_0 - (\mathbf{d}_0 \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}, \tag{21}$$

which is simply $\mathbf{d}_0$ minus its component along the streamline.

Now equate $\mathbf{d}_1 = -(\mathbf{d}_0 \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}$ to Eqn. (18), which shows that the near point lies at $\delta\theta' = -c/r'$ where $c = (\mathbf{d}_0 \cdot \hat{\mathbf{t}})/N = (d_r s_r + d_z s_z)/N^2$. So at the near point

$$\mathbf{d}_1 = -Nc\,\hat{\mathbf{t}} = -c(s_r \hat{\mathbf{r}} + \hat{\boldsymbol{\theta}} + s_z \hat{\mathbf{z}}). \tag{22}$$

Adding $\mathbf{d}_0$ to the above then yields $\mathbf{\Delta}$:

$$\mathbf{\Delta} = (d_r - cs_r)\hat{\mathbf{r}} - c\hat{\boldsymbol{\theta}} + (d_z - cs_z)\hat{\mathbf{z}} \tag{23a}$$

$$\Delta^2 = d_r^2 + d_z^2 - (Nc)^2. \tag{23b}$$

Inserting $\mathbf{\Delta}$ into Eqn. (14) then provides the acceleration that streamline a' exerts on the particle:

$$A_{g,r} = \frac{2G\lambda'(d_r - cs_r)}{\Delta^2} \tag{24a}$$

$$A_{g,\theta} = -\frac{2G\lambda' c}{\Delta^2} \tag{24b}$$

$$A_{g,z} = \frac{2G\lambda'(d_z - cs_z)}{\Delta^2}, \tag{24c}$$

where $N^2 = 1 + s_r^2 + s_z^2$ and $c = (d_r s_r + d_z s_z)/N^2$.

Lastly, note that because the streamline is tilted from $\hat{\boldsymbol{\theta}}$ by the angle φ, the right triangle of Eqn. (17) has $|\mathbf{d}_1| \le \Delta \tan\varphi = s\Delta$, with equality in the ring plane (Fig. 1). An m-armed disturbance displaces streamline a' radially by about ea' and vertically by about ia' where e and i are its particles' eccentricity and inclination, so Eqns. (19) give $s_r \sim me$ and $s_z \sim mi$, and $s \lesssim m(e^2 + i^2)^{1/2}$. Rings are nearly circular and nearly planar, and m is modest for the waves of interest, so $s \ll 1$ and $|\mathbf{d}_1| \ll \Delta$, which means that the near point lies very close to the crossing point.

3.1.2. *Computing the acceleration*

The particles' coordinates are stored in arrays whose rows are the streamlines, and every timestep, before the accelerations are computed, the particles in each row are sorted by longitude. Because the streamlines are nearly concentric, the particle in row a' that has the perturbed particle's column index lies at nearly the same longitude, so it and its neighbor on either side bracket θ. Lagrange polynomials fit to those three particles then provide, by interpolation to θ, the crossing point's $r'(\theta)$ and $z'(\theta)$ and hence d_r and d_z, the linear density $\lambda'(\theta)$, and, from the polynomials' derivatives, the slopes s_r and s_z of Eqn. (19), and Eqns. (23)–(24) then provide the acceleration. This is repeated for every perturbing streamline, and the sum is the particle's acceleration due to ring gravity. Note that λ' is evaluated at the crossing point rather than at the near point, which is justified since Eqn. (14) assumes that λ' varies insignificantly over distances of several Δ along the streamline, while the near point lies only $|\mathbf{d}_1| \ll \Delta$ from the crossing point.

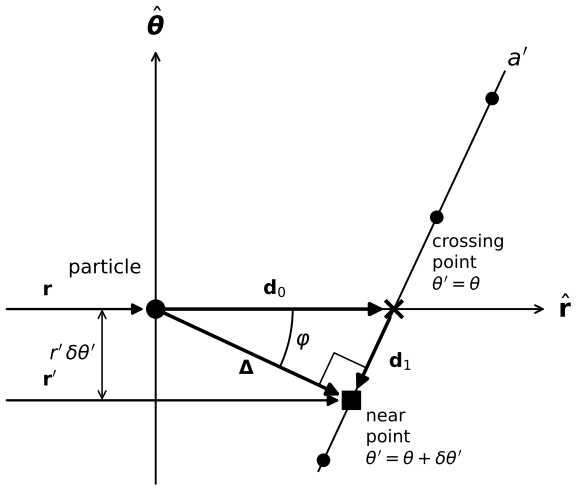


Figure 1. The gravity of a perturbing streamline a' on a particle in the ring plane: Fig. 3 of J. M. Hahn & J. N. Spitale (2013), redrawn. Near the particle's longitude, a' (thin line, with the three particles nearest in longitude) is straight, tilted from $\hat{\boldsymbol{\theta}}$ by φ where $\tan\varphi = s_r$, and it crosses that longitude at $\mathbf{d}_0 = d_r\hat{\mathbf{r}}$ from the particle (cross). Its near point (black square) lies $\mathbf{d}_1$ further along it, $r'\,\delta\theta'$ along $\hat{\boldsymbol{\theta}}$ from the crossing point (here $\delta\theta' < 0$). The perpendicular $\mathbf{\Delta} = \mathbf{r}' - \mathbf{r} = \mathbf{d}_0 + \mathbf{d}_1$ (Eqn. 15), where $\mathbf{r}$ and $\mathbf{r}'$ are the positions of the particle and the near point (the planet is far off to the left), meets the streamline at a right angle, has length $\Delta = d_r \cos\varphi$ and makes the angle φ with $\hat{\mathbf{r}}$. The wire's acceleration $\mathbf{A}_g = (2G\lambda'/\Delta)\hat{\mathbf{\Delta}}$ points along it (Eqns. A1).

3.2. *Pressure*

`Epi_int_lite` treats the ring as a dilute gas of colliding particles whose one-dimensional pressure is $p = c^2\sigma$ where c is the particles' velocity dispersion and σ is the ring's surface density (Section 2.2 of J. M. Hahn et al. 2025). Its treatment in three dimensions, and a test against the theory of spiral density waves in a non-gravitating, pressure-supported gas disk, will be completed later.

3.3. *Viscosity*

`Epi_int_3d` uses the shear and bulk viscosity of `epi_int_lite` (Section 2.2 of J. M. Hahn et al. 2025; Appendix A of J. M. Hahn & J. N. Spitale 2013), which act only in the ring plane. A vertical viscosity, which would damp bending waves, will be completed later. The bending-wave simulations of Section 4 have no viscosity or pressure.

3.4. *Advancing the particles*

`Epi_int_3d` advances the particles with the second-order symplectic kick-drift-kick scheme of `epi_int_lite`, which is that of the MERCURY N-body integrator (J. E. Chambers 1999); see Section 2.2 of J. M. Hahn et al. (2025). Each timestep Δt starts with a kick that changes each particle's velocity by the ring's accelerations, plus any satellite's (Section 4), applied over $\Delta t/2$. Then a drift advances the particle along its unperturbed orbit for Δt, and a second half kick ends the timestep. The drift converts coordinates to elements (Section 2.3), advances $M \to M + \kappa_a \Delta t$ and $\Omega \to \Omega + (\Omega_a - \nu_a)\Delta t$, and converts back, so it is exact for the unperturbed motion of Eqns. (7) and (11). Its vertical part is simply a rotation of $(z, v_z/\nu_a)$ through the angle $\Delta\theta - (\Omega_a - \nu_a)\Delta t$, where $\Delta\theta$ is the drift's change in the particle's longitude. Before and after each drift, small shifts of the particles' coordinates account for the planet's motion about the system's center of mass (J. E. Chambers 1999).

4. USING SPIRAL WAVE THEORY TO TEST `EPI_INT_3D`

Here we use `epi_int_3d` to simulate the spiral bending waves that a satellite launches at its vertical resonances in a planetary ring, which exercises the code's new vertical capability, and we compare them with linear wave theory to confirm the code. These waves test the vertical motion of Section 2 and the vertical ring gravity of Section 3.1. We will also simulate the spiral density waves that a satellite launches at its Lindblad resonances, to further validate the code's ability to simulate horizontal disturbances in a self-gravitating ring, and in a non-gravitating, pressure-supported gas disk those waves will also test the ring pressure of Section 3.2. Each test's notebook, which makes the comparisons shown here and others, is in the code's repository; the simulations themselves are not.

4.1. *Spiral bending waves*

4.1.1. *Free spiral bending waves*

A spiral bending wave displaces the ring vertically by $Z = \mathrm{Re}[h(r)e^{i(m\theta - \omega t)}]$, where the integer m and the frequency ω are constant and the complex amplitude $h \propto e^{i\int k\,dr}$ winds with radial wavenumber k. The wave's crests, where $m\theta + \int k\,dr - \omega t$ is constant, rotate with the pattern speed $\Omega_p = \omega/m$, prograde when $\Omega_p > 0$, and a ring particle orbiting at Ω_a sees the wave at the Doppler-shifted frequency $\omega_m = m(\Omega_a - \Omega_p)$. The wave obeys the dispersion relation (Eqn. (84) of F. H. Shu 1984)[2]

$$\omega_m^2 = \nu_a^2 + 2\pi G\sigma|k|, \tag{25}$$

where σ is the ring's surface density, whose self-gravity adds to the vertical restoring force (Appendix C). Since $\omega = m\Omega_a - \omega_m$, the wave's group velocity is

$$c_g = \frac{\partial\omega}{\partial k} = -\mathrm{sgn}(k)\frac{\pi G\sigma}{\omega_m} \tag{26}$$

(Eqn. (85) of F. H. Shu 1984). The dispersion relation fixes only $|k|$, and the group velocity's sign sets the sign of k. A wave is launched at a vertical resonance, where $\omega_m = \nu_a > 0$, and its angular momentum luminosity, which equals the torque that launches it, Eqn. (40), flows away from the resonance, into the region where $\omega_m^2 > \nu_a^2$. At an inner vertical resonance, where $m > 0$, that region lies inside the resonance, so $c_g < 0$ and $k > 0$; at an outer vertical resonance, where $m < 0$, it lies outside, so $c_g > 0$ and $k < 0$. Either way k has the sign of m, so along a crest the longitude θ decreases with r, and the wave is trailing (F. H. Shu 1984).

4.1.2. *Forced spiral bending waves*

A satellite of mass M_s whose circular orbit of radius a_s is inclined by i_s to the ring plane launches spiral bending waves at its vertical resonances (F. H. Shu et al. 1983; F. H. Shu 1984). To first order in i_s and z, the satellite accelerates a ring particle at radius r and longitude θ vertically by

$$A_z = -\frac{GM_s(z - z_s)}{\Delta_s^3} \tag{27}$$

where $z_s = a_s i_s \sin(\theta_s - \Omega_s)$ is the satellite's height above the ring plane, its longitude $\theta_s = \Omega_{a_s} t$ and longitude of ascending node $\Omega_s = (\Omega_{a_s} - \nu_{a_s})t$ advance at constant rates from zero at time $t = 0$, and Δ_s is the particle's distance from the satellite, where $\Delta_s^2 = r^2 + a_s^2 - 2ra_s\cos(\theta - \theta_s)$ omits $(z - z_s)^2$, which is of second order. The ring's vertical excursions are much smaller than the satellite's, $|z| \ll a_s i_s$, so $A_z \simeq$

[2] F. H. Shu (1984) writes perturbations as $e^{i(\omega t - m\theta)}$, the complex conjugate of the form used here. His equations are therefore the complex conjugates of those here, with k of opposite sign: his trailing waves at an inner vertical resonance have $k < 0$, where those here have $k > 0$.

$GM_s z_s/\Delta_s^3$, whose Fourier expansion in θ is

$$A_z = -\sum_{m=-\infty}^{\infty} A_m \sin[m\theta - (m\Omega_{a_s} + \nu_{a_s})t] \tag{28}$$

where

$$A_m = \frac{GM_s i_s}{2a_s^2} b_{3/2}^{(m)}(\beta) \tag{29}$$

in which

$$b_{3/2}^{(m)}(\beta) = \frac{2}{\pi}\int_0^\pi \frac{\cos m\phi\, d\phi}{(1+\beta^2-2\beta\cos\phi)^{3/2}} \tag{30}$$

is a Laplace coefficient of $\beta = r/a_s$ (Eqn. (A21) of F. H. Shu 1984). Each term with $|m| \geq 2$ is an $|m|$-armed pattern that rotates with angular velocity $\Omega_p = \Omega_{a_s} + \nu_{a_s}/m$ and launches a spiral bending wave at its one vertical resonance, which lies interior to the satellite's orbit when $m > 0$ and exterior when $m < 0$, and these resonances are well separated across the ring (F. H. Shu 1984). The $m = -1$ term drives a secular resonance where the ring's nodes regress at the satellite's rate, which is not considered here, and the $m = 0$ and $m = 1$ terms have no resonance. The simulations here follow particles near a single mth resonance, so `Epi_int_3d` keeps only the resonant term of the satellite's forcing,

$$A_z = -A_V \sin m(\theta - \Omega_p t), \tag{31}$$

where A_V is A_m evaluated at $\beta = r_V/a_s$. A particle feels the forcing at the Doppler-shifted forcing frequency ω_m of Section 4.1.1, and the inner vertical resonance lies where $\omega_m = \nu_a$, at the radius $r_V \simeq a_s[(m-1)/(m+1)]^{2/3}$ when J_2 is small. There the quantity

$$D \equiv \nu_a^2 - \omega_m^2, \tag{32}$$

a measure of the distance from resonance in frequency, vanishes.

Near r_V the ring's vertical displacement is $Z = \mathrm{Re}[h(x)e^{im(\theta-\Omega_p t)}]$ where $x = (r - r_V)/r_V$ is the fractional distance from resonance, and $D \simeq \mathcal{D}x$ where $\mathcal{D} = r\, dD/dr$ at r_V (Eqn. (87a) of F. H. Shu 1984), which is $3(m-1)\Omega_a^2$ when oblateness is negligible. The dispersion relation, Eqn. (25), is $D = -2\pi G\sigma|k|$, and the wave here is trailing, with $k > 0$. Acting on $h \propto e^{i\int k\, dr}$, k is $-i\, d/dr$, which turns the dispersion relation into the left side of Eqn. (88) of F. H. Shu (1984),

$$-i\epsilon\frac{dh}{dx} + xh = L_V, \tag{33}$$

whose right side is the satellite's forcing, where

$$\epsilon = \frac{2\pi G\sigma}{r_V \mathcal{D}} \quad \text{and} \quad L_V = \frac{iA_V}{\mathcal{D}}. \tag{34}$$

The solution of Eqn. (33) that has no wave at $x > 0$ is (Eqns. (89)–(90) of F. H. Shu 1984)

$$h(x) = -\frac{iL_V}{\epsilon} e^{-ix^2/2\epsilon} \int_x^\infty e^{is^2/2\epsilon}\, ds. \tag{35}$$

Far from resonance, where $|x| \gg \epsilon^{1/2}$, Eqn. (35) tends to Eqns. (91)–(92) of F. H. Shu (1984),

$$h = h_{\text{forced}} + h_{\text{wave}}, \tag{36a}$$

$$\text{where} \quad h_{\text{forced}} = \frac{L_V}{x}, \tag{36b}$$

$$\text{and} \quad h_{\text{wave}} = \begin{cases} Z_w e^{-ix^2/2\epsilon} & x \ll -\epsilon^{1/2} \\ 0 & x \gg \epsilon^{1/2}. \end{cases} \tag{36c}$$

The forced oscillation h_{forced}, which is iA_V/D to first order in x, is the response of a massless ring, and h_{wave} is a spiral bending wave. The wave's amplitude

$$|Z_w| = \left(\frac{2\pi}{\epsilon}\right)^{1/2} |L_V| = A_V \left(\frac{r_V}{G\sigma\mathcal{D}}\right)^{1/2} \tag{37}$$

is constant, while its wavelength

$$\lambda_{\text{wave}} = \frac{4\pi^2 G\sigma}{\mathcal{D}|x|} \tag{38}$$

shrinks with distance from resonance. The wave's first wavelength, from r_V to where its phase $x^2/2\epsilon$ reaches 2π, is $(4\pi\epsilon)^{1/2} r_V$. Appendix D shows that $h \to h_{\text{forced}}$ as $\sigma \to 0$.

Equations (33)–(38) give us a few opportunities to test `epi_int_3d`'s vertical motion. The forced oscillation alone is tested by a ring of massless particles (Section 4.1.3), which checks the forcing A_V, the resonance's location r_V and $\mathcal{D}$, which together also set the wave's amplitude. The wave is tested by a self-gravitating ring (Section 4.1.4). A third test may include the ring's viscosity, which damps the wave (Eqns. (97)–(98) of F. H. Shu 1984).

4.1.3. *Massless particles*

A ring of massless particles has no wave, only the forced oscillation h_{forced} of Eqn. (36), which, expressed as an inclination $|h_{\text{forced}}|/r_V$, is

$$i_f = \frac{M_s}{M_p}\frac{\beta^2 b_{3/2}^{(m)}(\beta)}{6(m-1)|x|}\, i_s \tag{39}$$

when the planet's oblateness is negligible. The simulation uses $J_2 = 0.01$ and $R_p = 0.5$, in natural units where $G = M_p = 1$. Its ring is 81 streamlines of 6 particles each, spanning $1 \leq r \leq 1.04$, and an $m = 2$ satellite with $i_s = 0.01$ at $a_s = 2.125$ puts $r_V = 1.02$ mid-ring.

The satellite's mass grows as $M_s = 10^{-6}(1 - e^{-t/\tau})$ with $\tau = 10^4$, or about 1600 orbits, slowly enough that the particles' motion is mostly forced rather than free, and what free motion remains averages away over time. Figure 2 shows that the particles' inclinations, averaged over the last 10^4 of the simulation's 2×10^4 orbits, are in excellent agreement with Eqn. (39), to within 4%, and the differences are of order x, which Eqn. (39) neglects. The test's notebook[3] also compares each particle's vertical motion over time to the exact solution, which the simulation follows to within 4×10^{-11} in z.

The forced response also locates the resonance. A particle forced by Eqn. (31) oscillates as $z = \mathrm{Re}[he^{im(\theta - \Omega_p t)}]$ where $\mathrm{Im}(h) = A_V/D$, so each streamline's amplitude h, fit to its particles' z and averaged over the last 10^4 orbits, gives the simulated $D = A_V/\mathrm{Im}(h)$. Figure 3 shows that it agrees with the exact D at the streamline's semi-major axis to within 0.4%, so it vanishes where theory puts the resonance, at r_V, and it departs from the linear approximation $\mathcal{D}x$ by terms of order x^2, as theory does.

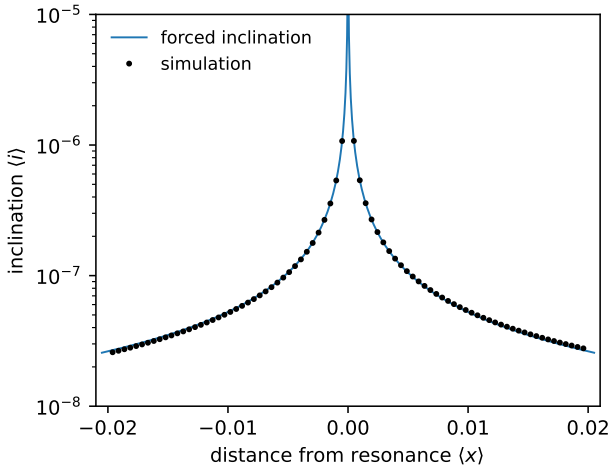


Figure 2. Inclinations $\langle i \rangle$ of massless particles near a satellite's $m = 2$ inner vertical resonance, averaged over each streamline's particles and over the last 10^4 orbits of the simulation of Section 4.1.3 (dots), plotted vs. their average fractional distance from resonance $\langle x \rangle$. The curve is the forced inclination, Eqn. (39). The streamline at $x = 0$, whose inclination grows without bound, is off scale.

4.1.4. *A self-gravitating ring*

The same kind of satellite also launches a spiral bending wave in a ring that has mass. This ring is 540 streamlines of 32 particles each, spanning $1 \le r \le 1.008$, whose

[3] https://github.com/joehahn/epi_int_3d/blob/main/tests/vertical_resonance/vertical_resonance.ipynb

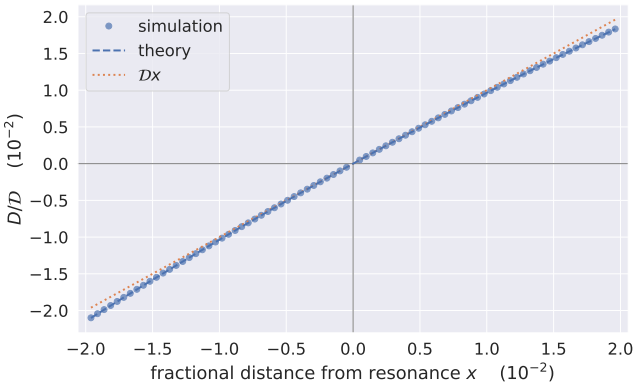


Figure 3. The quantity $D = \nu_a^2 - \omega_m^2$, which vanishes at the resonance, in units of $\mathcal{D}$, for each streamline of the massless ring of Section 4.1.3 (dots), inferred from its forced vertical amplitude averaged over the last 10^4 orbits, plotted vs. its fractional distance from resonance x. The dashed curve is the exact D, and the dotted line its linear approximation $\mathcal{D}x$. The streamline at $x = 0$, whose amplitude grows without bound, is left out.

total mass 4×10^{-9} gives it the uniform surface density $\sigma = 7.9 \times 10^{-8}$. It has no viscosity or pressure, and J_2 and R_p are as in Section 4.1.3. An $m = 2$ satellite of mass $M_s = 2.5 \times 10^{-7}$ and inclination $i_s = 0.01$ at $a_s = 2.097$ puts the resonance at $r_V = 1.0064$, 80% of the way across the ring, and the satellite is at full mass from $t = 0$. These parameters do not correspond to any real ring and satellite. Rather, they were chosen so that the wave is linear and well resolved: its amplitude $|Z_w|$, Eqn. (37), keeps $|kZ_w| \le 0.03$ everywhere in the ring, and its wavelength, Eqn. (38), spans about 11 streamlines even at the ring's inner edge, where it is shortest.

The wave propagates inward at the speed $|c_g| = \pi G\sigma/\nu_a$, Eqn. (26) near resonance, so it takes the time $\tau_c = (r_V - 1)/|c_g|$, about 4100 orbits, to cross from r_V to the ring's inner edge. The simulation stops at 3500 orbits, or $0.86\,\tau_c$, before the wave can reflect there. Figure 4 shows the wave being launched at r_V and propagating inward. Switching the satellite on at full mass also starts a free vertical oscillation everywhere, so at first every streamline moves as a massless particle would. The ring's self-gravity then takes over: the response near r_V settles onto linear theory, and the wave front advances inward at about $|c_g|$. The front overshoots linear theory a little, since it carries along the free oscillation that was started at resonance, and a small residue of that oscillation remains ahead of the front. Behind the front, though, the simulated inclinations are in excellent agreement with linear theory, with about 14 wavelengths of the wave in place by the end of the run. There the wave's amplitude is within 3% of $|Z_w|$, and its phase winds inward as the trailing wave of Eqn. (36) does. On

the nonwave side of the resonance, the forced response is up to 9% larger than linear theory's. That theory is for an infinite sheet, which has ring matter beyond this ring's outer edge, and that is the likely cause, though it has not been quantified. The satellite pulls only vertically, and the wave leaves no mark in the ring plane: no particle moves radially by more than 4×10^{-8}, well under 1% of the streamline spacing $\Delta = 1.5 \times 10^{-5}$.

Figure 5 shows the simulated wave's wavenumber $|k|$, measured from the radial gradient of the phase of its wave part $h - L_V/x$, as in Fig. 5 of J. M. Hahn (2007). Behind the front it is 5.5% larger than the $|k| = 2\pi/\lambda_{\text{wave}}$ of Eqn. (38) (the median), i.e., its wavelength is that much shorter. This is because the simulated ring is a finite lattice of streamlines rather than the infinite, continuous sheet that Eqn. (38) assumes, and both differences weaken the wave's restoring force. Appendix C shows that the lattice's spacing reduces self-gravity's restoring force from $2\pi G\sigma|k|$ to $X(k\Delta)$, and that the ring's edges reduce it further, by $2G\sigma(1/d_{\text{in}} + 1/d_{\text{out}})$. With the lattice allowed for, the simulated wavenumber is 2.3% larger than expected, and with the edges allowed for too, Eqn. (C3), it is 0.9% larger. All 187 of the 187 wavenumbers measured behind the front are positive, so the wave is trailing, as Section 4.1.1 requires. Its phase there, fit against time, advances at the forcing's pattern speed Ω_p to within a fraction 6.3×10^{-5}, so it rotates prograde with the forcing. And the first wavelength that its phase implies is 2.9% shorter than $(4\pi\epsilon)^{1/2} r_V$, for the same reasons as its wavenumber is larger. The test's notebook[4] also compares the wave's complex amplitude with linear theory.

The torque that the satellite exerts on the wave tests the wave's amplitude and phase together. The satellite's resonant potential $A_V z \sin m(\theta - \Omega_p t)$, whose vertical pull is Eqn. (31), exerts the torque $T = -mA_V \sum \mu z \cos m(\theta - \Omega_p t)$ on the simulated ring, summed over its particles of mass μ, while linear theory gives (F. H. Shu 1984)

$$T_V = -\frac{\pi^2 m \sigma r_V^2 A_V^2}{\mathcal{D}}, \tag{40}$$

which is also the angular momentum luminosity of the wave that the satellite launches. Figure 6 shows that T settles onto T_V as soon as the wave is launched. Over the run's last half its mean is $0.996\,T_V$, and it scatters by at most 11% about T_V because of the residue of the free oscillation. The satellite pulls only vertically, so the ring does not feel this torque, and its total angular momentum L_z can change only through the ring's gravity and the integrator's round-off error. Figure 7 shows that $|\Delta L_z/L_z| \le 1.4 \times 10^{-13}$ over the run.

[4] https://github.com/joehahn/epi_int_3d/blob/main/tests/bending_waves/bending_waves.ipynb

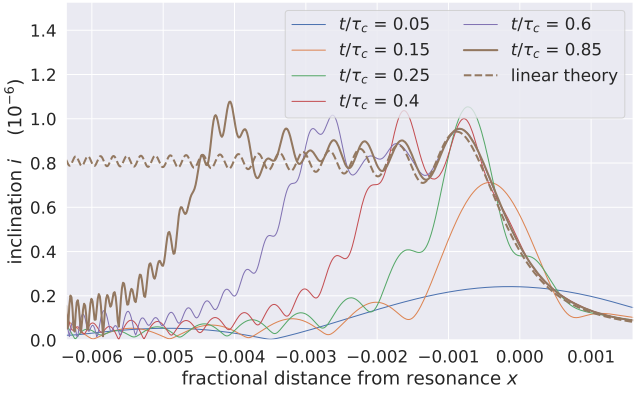


Figure 4. Inclinations i of the streamlines in the self-gravitating ring of Section 4.1.4, averaged over each streamline's particles, plotted vs. their fractional distance from resonance x, at six times t in units of the time τ_c that the wave takes to propagate from r_V to the ring's inner edge, which is at the left. The dashed curve is linear theory's inclination $|h|/r$, with h from Eqn. (35). An animation of the wave is at joehahn.github.io/epi_int_3d.

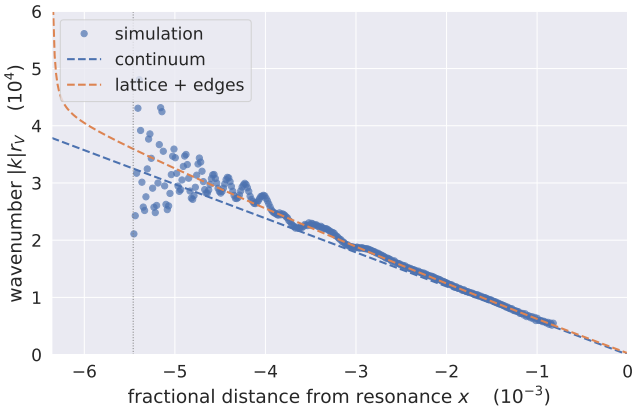


Figure 5. The wavenumber $|k|r_V$ of the bending wave in the self-gravitating ring of Section 4.1.4 at the end of the run (dots), plotted vs. fractional distance from resonance x, from the front (dotted line) to the resonance zone. The blue dashed line is the continuum's dispersion relation, Eqn. (38), and the orange dashed curve that of the finite lattice of streamlines with the ring's edges, Eqn. (C3).

4.2. *Spiral density waves*

Spiral density waves launched at a satellite's Lindblad resonance, in a self-gravitating ring and in a non-gravitating, pressure-supported gas disk, will be compared with linear theory later.

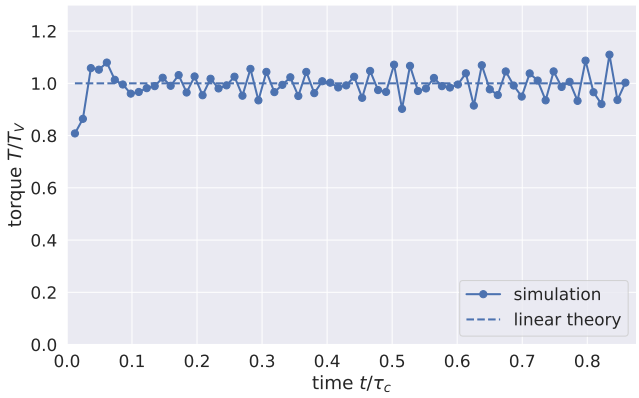


Figure 6. The torque T that the satellite exerts on the bending wave of Section 4.1.4, in units of linear theory's T_V, Eqn. (40) (dashed), plotted vs. time t in units of the wave's crossing time τ_c.

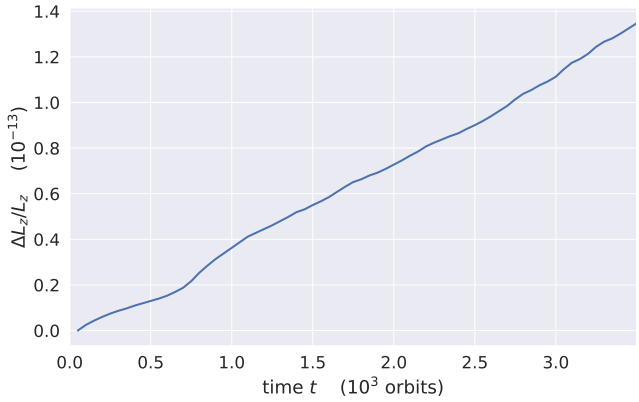


Figure 7. The fractional change in the total angular momentum L_z of the self-gravitating ring of Section 4.1.4 vs. time t in orbits.

APPENDIX

A. RELATION TO `EPI_INT` AND `EPI_INT_LITE`

In the ring plane $d_z = s_z = 0$, and the acceleration of Section 3.1.1 reduces to the geometry of Fig. 3 of J. M. Hahn & J. N. Spitale (2013), which is redrawn here as Fig. 1. There the streamline's tilt φ from $\hat{\boldsymbol{\theta}}$ has $\tan\varphi = s_r$. Then $c = d_r \sin\varphi\cos\varphi$ and $\boldsymbol{\Delta} = d_r\cos\varphi(\cos\varphi\hat{\mathbf{r}} - \sin\varphi\hat{\boldsymbol{\theta}})$, i.e., the perpendicular has length $\Delta = d_r\cos\varphi$ and makes the angle φ with $\hat{\mathbf{r}}$. The acceleration is then

$$A_{g,r} = \frac{2G\lambda'}{d_r} \tag{A1a}$$

$$A_{g,\theta} = -\frac{2G\lambda'\tan\varphi}{d_r}. \tag{A1b}$$

Out of the plane, the tangential acceleration $A_{g,\theta} = -2G\lambda' c/\Delta^2$ of Eqns. (24) also includes the term $-2G\lambda' d_z s_z/N^2\Delta^2$, which is due to the streamline's vertical tilt, and this is how vertically displaced streamlines exchange angular momentum.

J. M. Hahn & J. N. Spitale (2013) give the components of this acceleration as $A_{g,r} = A_g\cos\varphi$ and $A_{g,\theta} = -A_g\sin\varphi$, with $A_g = 2G\lambda/\Delta$ and Δ the perpendicular distance, and approximate them as $A_{g,r} \simeq A_g$ and $A_{g,\theta} \simeq -A_g v_r'/v_\theta'$ (their Eqns. 20), to lowest order in the streamline's eccentricity. With Δ read as the radial gap d_r and $\tan\varphi = v_r'/v_\theta'$, those are Eqns. (A1) without approximation. `Epi_int_lite` (J. M. Hahn et al. 2025) differs from Eqns. (A1) in three ways, which `epi_int_3d` corrects.

Distance—`Epi_int_lite` evaluates the pull's magnitude at the crossing point, $\Delta = d_r$, but keeps its direction toward the near point,

$$A_{g,r} = \frac{2G\lambda\cos\varphi}{d_r} \tag{A2a}$$

$$A_{g,\theta} = -\frac{2G\lambda\sin\varphi}{d_r}. \tag{A2b}$$

This pull is weaker than Eqns. (A1) by the factor $\cos\varphi = 1 - \mathcal{O}(\varphi^2)$. Which is why `epi_int` and `epi_int_lite` could use the distance to the crossing point, $|\mathbf{d}_0|$, for Δ, also out of the plane, i.e., neglect $\mathbf{d}_1$ in the pull's magnitude. Section 3.1.1 shows that the near point lies very close to the crossing point, $|\mathbf{d}_1| \le s\Delta$ with $s \lesssim m(e^2+i^2)^{1/2}$, and inserting that into $|\mathbf{d}_0|^2 = \Delta^2 + |\mathbf{d}_1|^2$ (Eqn. 17) gives $|\mathbf{d}_0|/N \le \Delta \le |\mathbf{d}_0|$. So the two differ by a fraction of order $s^2 \lesssim m^2(e^2+i^2)$, which is $\lesssim 10^{-4}$ even for $m(e^2+i^2)^{1/2}$ as large as 10^{-2} and so of no practical consequence, but Eqns. (A1) cost no more.

Direction—`Epi_int_lite` takes the wire's direction from the streamline's velocity at the crossing point, $\sin\varphi = v_r'/v'$ and $\cos\varphi = v_\theta'/v'$ (Fig. 8). The velocity is tangent to the streamline only when the streamline's shape is stationary. If the shape is a pattern that rotates at Ω_p while its particles orbit at Ω_a, then $v_r'/v_\theta' = (1 - \Omega_p/\Omega_a)s_r$ where s_r is the slope of the streamline's shape (Eqn. 19). Near an m-armed Lindblad resonance $1 - \Omega_p/\Omega_a \simeq \pm 1/m$, so the velocity underestimates the wire's tilt, and with it the tangential pull that transports angular momentum between streamlines, by a factor of about m. For the slowly precessing $m = 1$ ringlets of J. M. Hahn et al. (2025), Ω_p is the precession rate and the error is negligible. `Epi_int_3d` takes the direction from the shape (Eqn. 20).

Linear density—`Epi_int_lite` evaluates every perturbing streamline's pull with the perturbed particle's own linear density λ, rather than the perturbing streamline's λ'. The difference, $\lambda' - \lambda \simeq (a' - a)\partial\lambda/\partial a$, adds a nearly uniform pull of order the ring's mass over the planet's; in the eccentric ringlets of J. M. Hahn et al. (2025) it changes the precession rate by about 10^{-7} of itself.

In an eccentric ringlet the three corrections together change the streamlines' eccentricities by less than 10^{-4} of themselves over 5000 orbits. Only the second matters for spiral waves, where it changes the torque between streamlines by a factor of about m. As in `epi_int_lite`, the particle's own streamline exerts no force on it in `epi_int_3d`, and Δ never vanishes, since streamlines may not cross. The slopes require that the streamline be well sampled by its

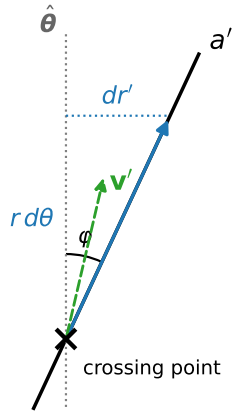


Figure 8. The perturbing streamline at its crossing point (Fig. 1). Its shape has the slope $\tan\varphi = s_r$: a step $r\,d\theta$ along $\hat{\boldsymbol{\theta}}$ moves it dr' outward. Its velocity $\mathbf{v}'$ (dashed), which `epi_int_lite` takes as the wire's direction, is tilted less when the shape rotates, here by half as much, as for an $m = 2$ pattern.

particles. `Epi_int_3d` also keeps `epi_int_lite`'s ring gravity, Eqns. (A2), as an option, so that its results can be checked against `epi_int_lite`'s exactly.

B. RESOLUTION LIMIT OF THE STREAMLINE METHOD

A streamline model represents a continuous ring by N_r streamlines spaced Δ apart, and that discreteness sets which wavelengths of the ring's response it can follow. This appendix derives the axisymmetric stability of a uniform lattice of circular streamlines under the self-gravity and pressure accelerations of `epi_int_lite` (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025), and compares it with the continuum Toomre criterion.

Each streamline of linear density λ exerts the acceleration $2G\lambda/d$ of a straight wire on a particle a distance d away (J. M. Hahn & J. N. Spitale 2013). The pressure $p = c^2\sigma$ acts through the surface density $\sigma_j = 2\lambda/(r_{j+1} - r_{j-1})$ of streamline j and the central difference $A_j = -(p_{j+1} - p_{j-1})/2\lambda$, which tends to $-\sigma^{-1}\partial p/\partial r$ as $\Delta \to 0$. Displace the streamlines radially by $\xi_j = \xi\cos(j\phi)$ where $\phi = k\Delta$ and k is the radial wavenumber, so that $0 < \phi \le \pi$; $\phi = \pi$ is the shortest wave the lattice can carry, the zigzag $\xi_j = (-1)^j\xi$ of Fig. 9a. Linearizing in ξ, the streamlines at $j = \pm n$ add $4G\lambda(1 - \cos n\phi)\xi/(n\Delta)^2$ to the acceleration of streamline 0, and summing over n with $\sigma = \lambda/\Delta$ gives

$$X(\phi) = \frac{4G\sigma}{\Delta}\sum_{n=1}^{\infty}\frac{1-\cos n\phi}{n^2} = \frac{4G\sigma}{\Delta}\left(\frac{\pi\phi}{2} - \frac{\phi^2}{4}\right), \tag{B1}$$

while pressure contributes

$$\omega_p^2(\phi) = \left(\frac{c}{\Delta}\right)^2 \sin^2\phi, \tag{B2}$$

so that each mode oscillates at the frequency

$$\omega^2 = \kappa^2 - X(\phi) + \omega_p^2(\phi). \tag{B3}$$

For $\phi \ll 1$ these recover the continuum terms $X \to 2\pi G\sigma k$ and $\omega_p^2 \to c^2k^2$, and Eqn. (B3) becomes the dispersion relation of a pressure-supported self-gravitating sheet, which is stable at all k when $Q = c\kappa/\pi G\sigma \ge 1$ (A. Toomre 1964; J. Binney & S. Tremaine 1987). At the grid scale the lattice departs from the continuum in two ways (Fig. 9b). Self-gravity peaks at $X(\pi) = \pi^2 G\sigma/\Delta$, half the continuum value $2\pi G\sigma k$ at $k = \pi/\Delta$. And pressure vanishes there: in the zigzag, streamlines $j \pm 1$ move together, so $r_{j+1} - r_{j-1}$ and hence σ_j and p_j do not change. The viscous fluxes of J. M. Hahn & J. N. Spitale (2013), which are differenced in the same way, likewise exert no force on the zigzag.

A cold continuum disk ($c = 0$) is unstable to all wavelengths shorter than $\lambda_{\rm crit} = 4\pi^2 G\sigma/\kappa^2$, and the lattice hides these: by Eqn. (B1) a cold lattice is axisymmetrically stable as long as

$$\mathcal{R} \equiv \frac{\lambda_{\rm crit}}{2\Delta} = \frac{2\pi^2 G\sigma}{\kappa^2\Delta} \simeq \frac{\pi G m_r N_r}{\kappa^2 a\Delta a^2} < 2, \tag{B4}$$

where 2Δ is the shortest resolved wavelength, and the last form is for a ringlet of mass m_r, semimajor axis a and width Δa, with $\sigma \simeq m_r/2\pi a\Delta a$ and $\Delta \simeq \Delta a/N_r$. Adding streamlines to a given ringlet raises $\mathcal{R}$ in proportion to N_r.

A viscous ringlet must also avoid the viscous overstability, the growth of axisymmetric epicyclic oscillations that viscous stresses drive when the shear stress rises steeply enough with surface density (U. Schmit & W. M. Tscharnuter 1995; P.-Y. Longaretti & N. Rappaport 1995; H. Salo et al. 2001; J. Schmidt et al. 2001; H. N. Latter & G. I. Ogilvie 2006, 2008). For a cold ringlet with constant kinematic shear and bulk viscosities ν_s and ν_b, as in `epi_int_lite`, and Keplerian $\kappa = \Omega$, a mode with oscillation frequency $\omega \simeq \kappa(1-x)^{1/2}$ grows to first order in the viscosities at the rate

$$\gamma = \frac{K}{6(1-x)}\left[\nu_s(4x+2) - 3\nu_b(1-x)\right], \tag{B5}$$

where $K = k^2$ and $x = 2\pi G\sigma k/\kappa^2$ in the continuum. Without self-gravity ($x = 0$) the ringlet is stable when $\nu_b > 2\nu_s/3$, the onset found by U. Schmit & W. M. Tscharnuter (1995) and M. Lehmann et al. (2017) for a viscosity independent of σ, while self-gravity is destabilizing (M. Lehmann et al. 2017): a mode is stable only for $x < (3\nu_b - 2\nu_s)/(4\nu_s + 3\nu_b)$, which is 1/7 for $\nu_b = \nu_s$ and 0.4 for $\nu_b = 2\nu_s$. The streamline lattice follows Eqn. (B5) with the grid's versions of both terms: $K = \sin^2\phi/\Delta^2$, from the code's central differences of the viscous stresses, and $x = X(\phi)/\kappa^2$. Both stresses vanish at the zigzag, $\phi = \pi$, where self-gravity is strongest, $X(\pi)/\kappa^2 = \mathcal{R}/2$; so a viscous lattice is stable at every wavelength when $\mathcal{R} < 2/7$ for $\nu_b = \nu_s$, or $\mathcal{R} < 4/5$ for $\nu_b = 2\nu_s$, more stringent than Eqn. (B4). Since $K\nu \sim \nu/\Delta^2$, an overstable lattice's modes grow faster the more streamlines it has.

That limit matters because a ringlet's self-gravity must be strong enough to resist the differential precession due to J_2. Across a ringlet of width Δa, J_2 drives a differential precession $\propto J_2(R_p/a)^2\Omega_a\Delta a/a$, while for a small nonlinearity parameter the ringlet's self-gravity drives one $\propto (m_r/M_p)\Omega_a\Delta\tilde{\omega}(a/\Delta a)^2$, where $\Delta\tilde{\omega}$ is the periapse twist across it (cf. Eqn. C1 of J. M. Hahn et al. 2025). So keeping J_2 from winding up the ringlet requires $m_r \propto J_2\Delta a^3/\Delta\tilde{\omega}$, and then $\mathcal{R} \propto J_2\Delta a N_r/\Delta\tilde{\omega}$: a wider ringlet does not help, and for a given planet more streamlines can be afforded only by a narrower ringlet. This is tested here with simulations of a ringlet of width $\Delta a = 0.0125a$ whose edges are tapered, mass $m_r = 3\times10^{-8}M_p$, eccentricity $e = 10^{-4}$, periapses twisted by π from its inner to outer edge, and kinematic viscosity $\nu = 7\times10^{-11}a^2\Omega_a$, with no pressure, $R_p = 0.5a$ and $J_2 = 0.01$. With $N_r = 100$ streamlines ($\mathcal{R} \simeq 0.09$ in the ringlet's middle, where the taper makes σ 1.25 times and Δ 0.8 times their ringlet-wide means) the ringlet stays smooth for 10^4 orbits while self-gravity, rather than J_2, controls its shape: within about 1000 orbits it straightens the initial twist, and the ringlet's body then stays nearly aligned, as two sub-ringlets that precess slowly relative to each other. With $N_r = 300$ ($\mathcal{R} \simeq 0.3$) and $N_r = 1000$ ($\mathcal{R} \simeq 0.9$) it becomes ragged and its streamlines cross after about 190 and 30 orbits. At $N_r = 1000$ the disturbance starts in the ringlet's interior, grows at the same rate with 40 or 241 particles per streamline, and does not grow when ring gravity is turned off, consistent with the self-gravity-assisted overstability for $\nu_b = \nu_s$ at $\mathcal{R} > 2/7$. Pressure with $Q \le 2$ delays the crossing by less than two orbits, as expected from Fig. 9b, where $\omega_p^2 \le 0.1\kappa^2$.

The crossings near $\mathcal{R} \simeq 0.3$ are mostly a different, numerical, problem. The same ringlet without the initial twist, at $N_r = 333$ ($\mathcal{R} \simeq 0.31$), crosses after 145 orbits, and its code's own linearized axisymmetric modes, computed from the reference implementation's accelerations, grow by only 3.3×10^{-3} per orbit, far too slowly. The disturbance that crosses is instead a sawtooth along each streamline with azimuthal wavenumber $m \simeq N_t/2$ where N_t is the number of particles per streamline. It also grows in a circular ringlet, without ring gravity, and without viscosity, though more slowly (after about 220, 170 and 850 orbits), its onset varies non-monotonically with N_t (with $N_t = 30$ or 50 the ringlet survives 10^4 orbits, with 20, 40 or 320 it does not), and pressure with $Q \le 10$ delays it by at most 20%. It is presumably seeded by the 3-point Lagrange interpolation in longitude that couples neighboring streamlines' particles, and it remains an open problem.

So Eqn. (B4) is a necessary condition only. In practice a simulated ringlet whose self-gravity matters should be run with $\mathcal{R} \lesssim 0.1$, viscous or not, and a ringlet that needs more streamlines must be made narrower and much less massive.

C. THE RESTORING FORCE OF A BENDING WAVE

Equation (14) recovers the self-gravity of a bending wave. Consider a sheet of surface density σ made of streamlines with $\lambda = \sigma\, dr'$, displaced vertically to $Z(r) = Z_0\cos kr$. When $|d_z| \ll |d_r|$ and the slopes are small, $A_{g,z} \simeq 2G\lambda d_z/d_r^2$, and the sum of $A_{g,z}$ over the sheet is

$$A_z = 2G\sigma Z_0\int_{-\infty}^{\infty}\frac{\cos kx - 1}{x^2}\,dx = -2\pi G\sigma|k|Z, \tag{C1}$$

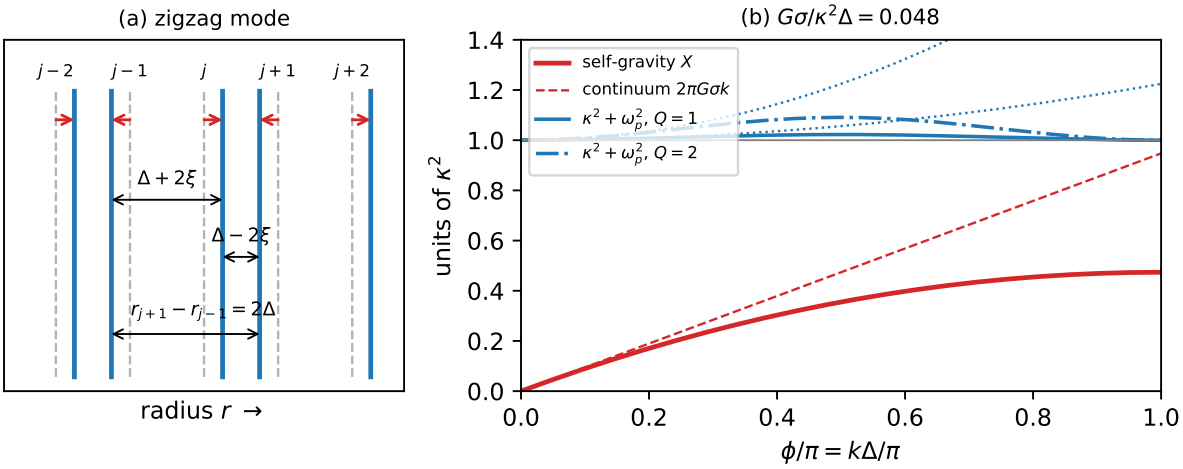


Figure 9. (a) The zigzag mode of a lattice of streamlines (blue) displaced by $\pm\xi$ from their unperturbed radii (dashed): the gaps alternate between $\Delta \pm 2\xi$, but $r_{j+1} - r_{j-1} = 2\Delta$ does not change, so neither does the surface density that sets streamline j's pressure. (b) The terms of Eqn. (B3) versus $\phi = k\Delta$, in units of κ^2, for $G\sigma/\kappa^2\Delta = 0.048$ (the ringlet of the text with $N_r = 1000$): the destabilizing self-gravity X (Eqn. B1, solid red) and its continuum value $2\pi G\sigma k$ (dashed red), and the restoring $\kappa^2 + \omega_p^2$ for $Q = 1$ and 2 (blue; Eqn. B2) and their continuum values $\kappa^2 + c^2k^2$ (dotted blue). A mode is unstable where X exceeds $\kappa^2 + \omega_p^2$.

which is the self-gravity term of the bending-wave dispersion relation, Eqn. (25) (F. H. Shu et al. 1983; F. H. Shu 1984). So summing the wires' pulls over the streamlines gives the vertical restoring force of a self-gravitating sheet, just as the same sum gives the $-2\pi G\sigma|k|$ term of a density wave in the plane. For bending waves, though, self-gravity is restoring, and adds to ν_a^2.

On the code's lattice of streamlines spaced Δ apart (here Δ is the spacing, as in Appendix B), with $Z_j = Z_0\cos(j\phi)$ and $\phi = k\Delta$,

$$A_z = \frac{4G\sigma Z_0}{\Delta}\sum_{n=1}^{\infty}\frac{\cos n\phi - 1}{n^2} = -X(\phi)Z_0, \tag{C2}$$

where $X(\phi)$ is the lattice self-gravity of Eqn. (B1). The vertical lattice modes therefore oscillate at $\omega^2 = \nu_a^2 + X(\phi)$, which tends to the continuum value as $\phi \to 0$. At the zigzag $\phi = \pi$, the extra restoring force is half the continuum value, as it is in the plane. Vertically, however, the lattice's self-gravity is stabilizing, so the resolution limit of Eqn. (B4), which concerns in-plane modes, has no vertical counterpart.

A ring of finite width also lacks the streamlines beyond its edges. Consider a streamline whose Nth neighbor on one side would be the first one beyond the ring's edge. The sum of Eqn. (C2) then stops at $n = N - 1$ on that side, and the missing terms are $(2G\sigma Z_0/\Delta)\sum_{n\geq N}(\cos n\phi - 1)/n^2$. Their cosines nearly cancel when the wave is shorter than the distance to the edge, while $\sum_{n\geq N} 1/n^2 \simeq 1/(N - \frac{1}{2})$, so the missing pull is about $2G\sigma Z_0/d$ where $d = (N - \frac{1}{2})\Delta$ is the streamline's distance to the ring's edge plus half a spacing. Near r_V, where $-D \simeq \mathcal{D}|x|$ on the wave side, the lattice's restoring force so replaces the continuum's $\mathcal{D}|x| = 2\pi G\sigma|k|$ with

$$\mathcal{D}|x| = X(k\Delta) - 2G\sigma\left(\frac{1}{d_{\rm in}} + \frac{1}{d_{\rm out}}\right), \tag{C3}$$

where $d_{\rm in}$ and $d_{\rm out}$ are the distances to the ring's inner and outer edges, so measured. Both terms lower the restoring force at a given k, so the wave's wavenumber is larger than in the continuum, by the amounts that Section 4.1.4 finds.

D. THE MASSLESS LIMIT OF THE BENDING WAVE

Equation (35) tends to the forced oscillation $h_{\rm forced} = L_V/x$ as the ring's surface density $\sigma \to 0$, which sends $\epsilon \to 0$. On the outer side of the resonance, $x > 0$, integrate the integral in Eqn. (35) by parts using $e^{is^2/2\epsilon} = (\epsilon/is)\,d(e^{is^2/2\epsilon})/ds$:

$$\int_x^\infty e^{is^2/2\epsilon}\,ds = -\frac{\epsilon}{ix}e^{ix^2/2\epsilon} + \frac{\epsilon}{i}\int_x^\infty \frac{e^{is^2/2\epsilon}}{s^2}\,ds. \tag{D1}$$

Integrating the last term by parts the same way shows that it is of order ϵ^2/x^3, smaller than the first by a factor of order ϵ/x^2. Inserting the first term into Eqn. (35) then gives

$$h = \frac{L_V}{x}\left[1 + \mathcal{O}\left(\frac{\epsilon}{x^2}\right)\right], \tag{D2}$$

so $h \to h_{\rm forced}$ as $\sigma \to 0$ at any fixed $x > 0$.

On the inner side, $x < 0$, the range of integration passes through $s = 0$, where ϵ/is diverges, so write the integral as $\int_{-\infty}^{\infty} - \int_{-\infty}^{x}$. The first is the Fresnel integral $(2\pi\epsilon)^{1/2}e^{i\pi/4}$, and the second integrates by parts as above, so

$$h = \frac{L_V}{x}\left[1 + \mathcal{O}\left(\frac{\epsilon}{x^2}\right)\right] - ie^{i\pi/4}\left(\frac{2\pi}{\epsilon}\right)^{1/2} L_V e^{-ix^2/2\epsilon}. \tag{D3}$$

The second term is $h_{\rm wave}$ of Eqn. (36c), with the amplitude $|Z_w|$ of Eqn. (37), and it grows as $\sigma^{-1/2}$ as $\sigma \to 0$. This is because Eqn. (35) is a steady state: the wave has had unlimited time to fill the ring inside r_V, and the satellite's torque on it, Eqn. (40), which is proportional to σ, is carried away at the speed $|c_g| = \pi G\sigma/\nu_a$, which also is proportional to σ. So the wave's angular momentum per unit area, which is proportional to $\sigma|Z_w|^2$, does not depend on σ. At any finite time t, though, the wave has reached only $x \simeq -|c_g|t/r_V$, which tends to zero with σ, so the inner side too has only the forced oscillation $h_{\rm forced}$ as $\sigma \to 0$. The massless ring of Section 4.1.3 is this limit.

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